

Assembly Instructions for the SpaceMount Guider Scope



Preface

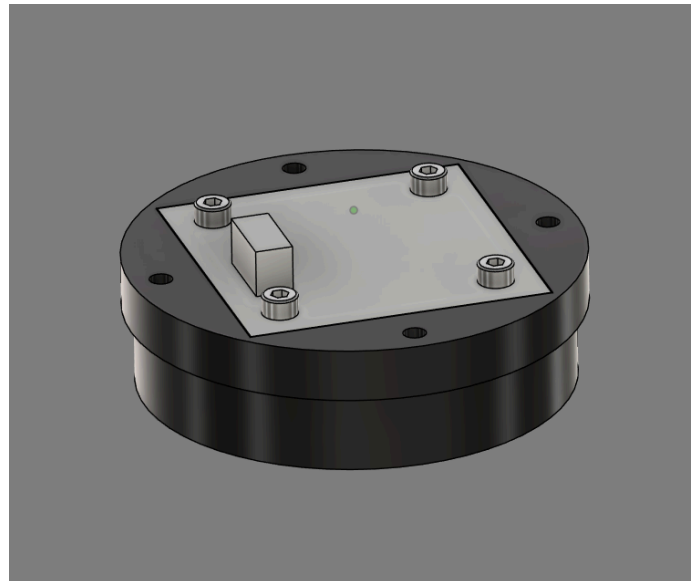
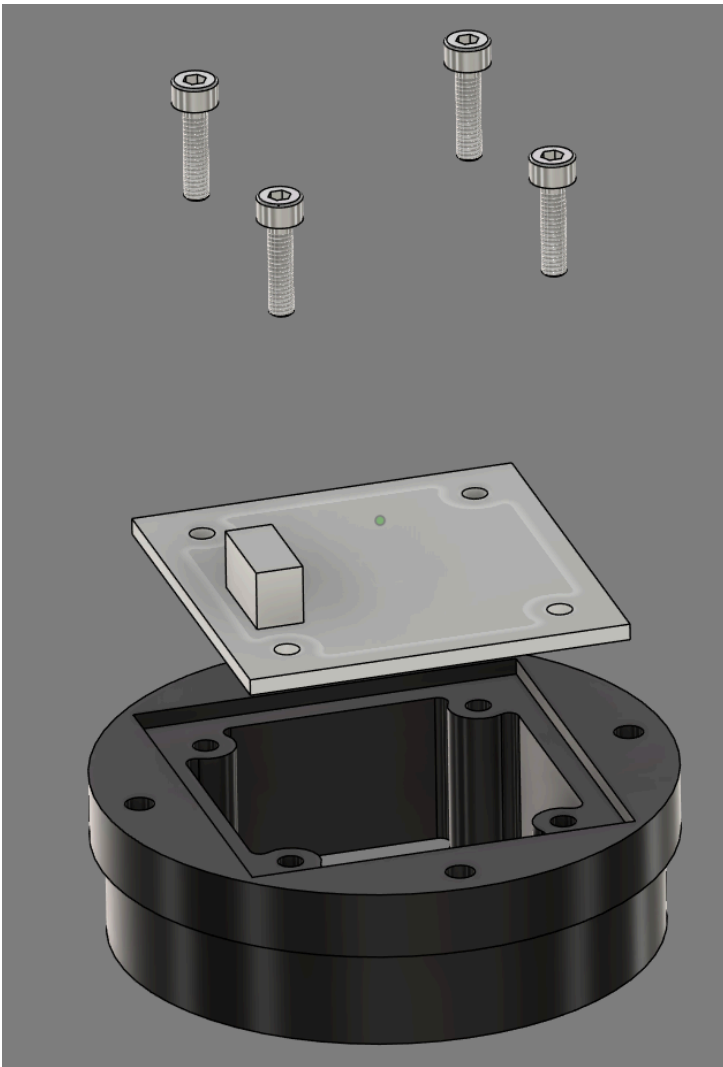
This is the assembly guide for my Guider Scope that is part of my [SpaceMount](#) project. It is certainly not error free, but it is written to the best of my abilities. Always refer to the master CAD assembly provided in the Github Repository for confirmation of assembly. Best of Luck!!

Note: Unless otherwise stated, assume screws are Socket Head Cap Screws (SHCS).

Part 1: Camera Module

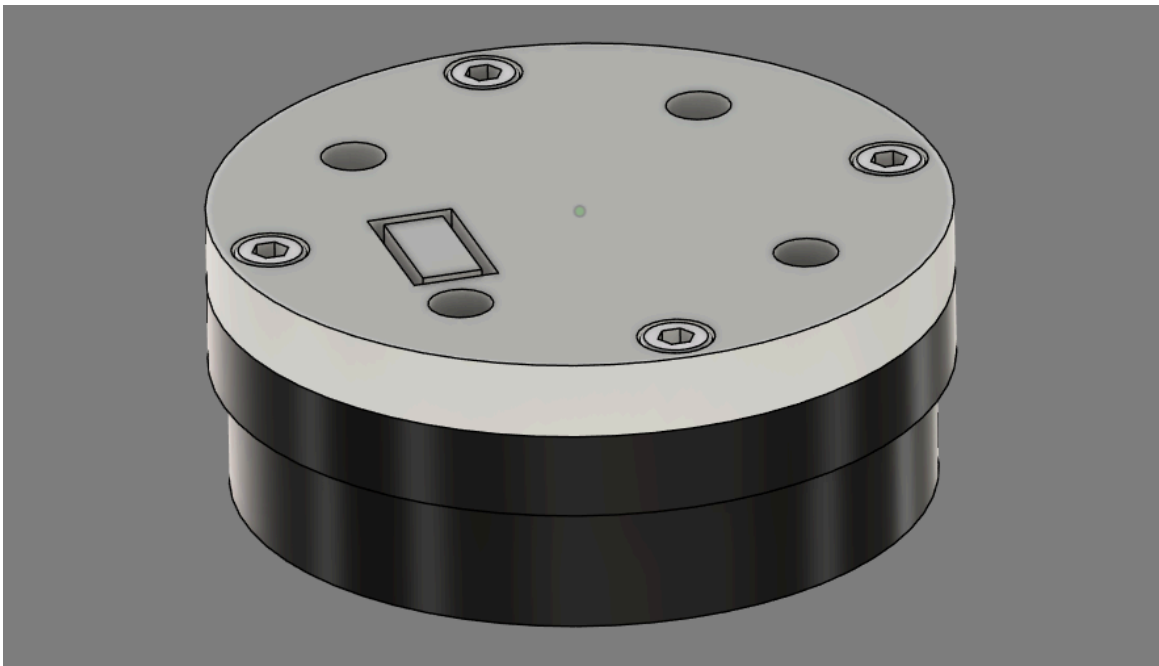
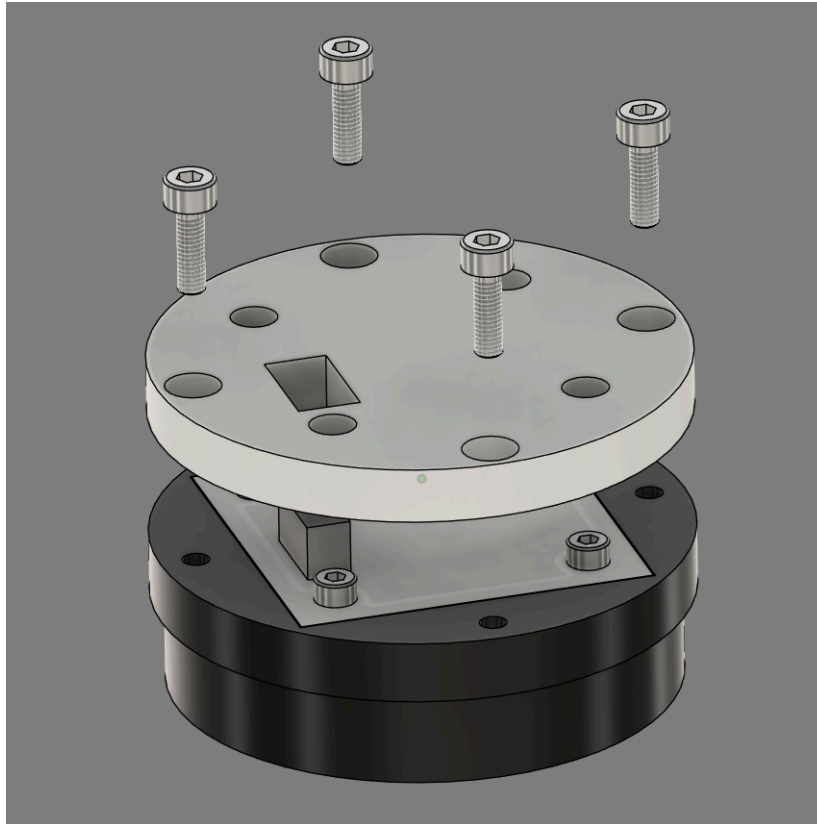
Step 1: Install Camera into Mount

- The IMX 290 camera module mounts into the Camera Mount using 4x M2.5x10mm Bolts

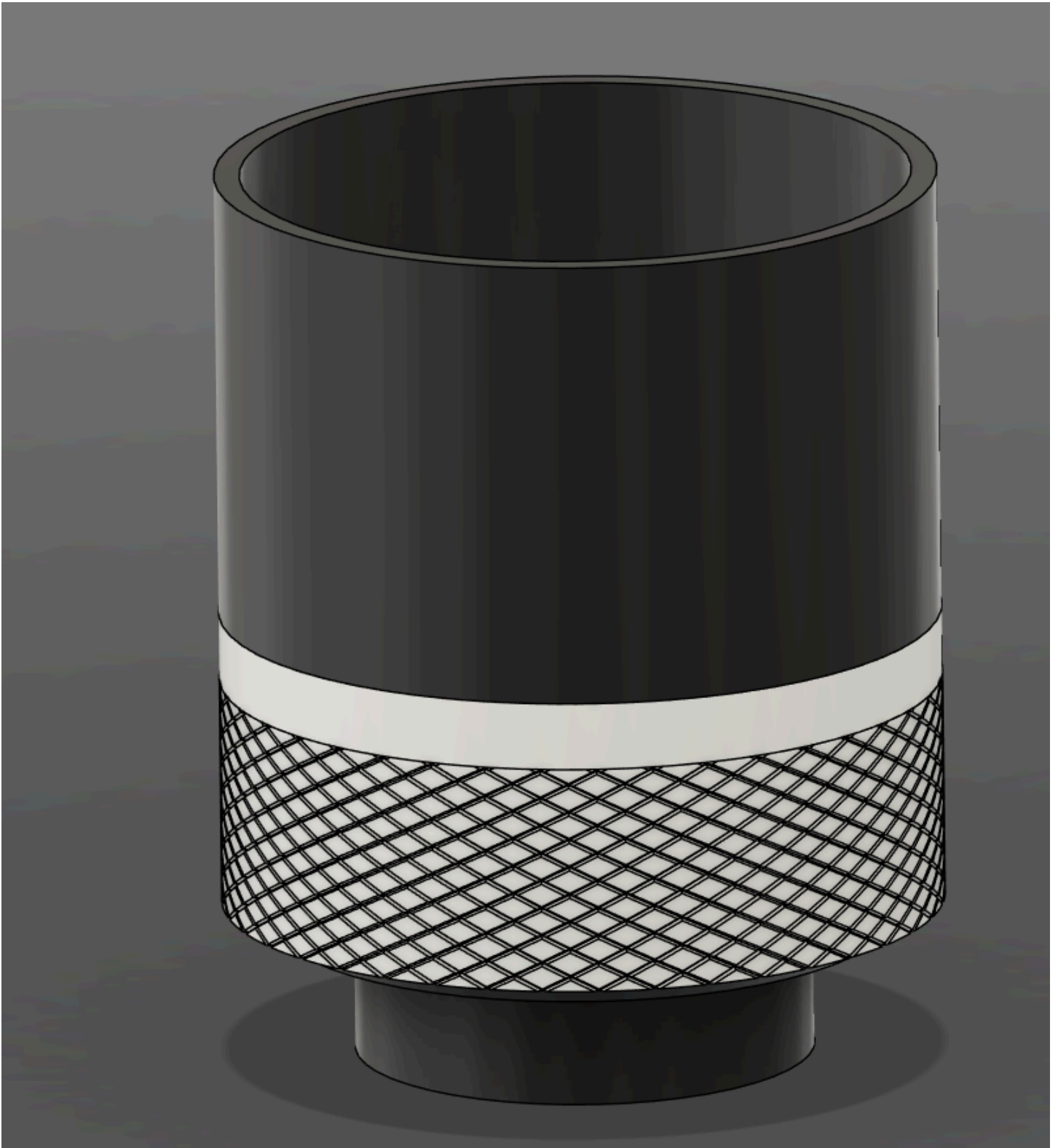


Step 2: Install Camera Cover

- The IMX290 Camera Cover screws into the Camera Mount with 4x M3x10mm Bolts.

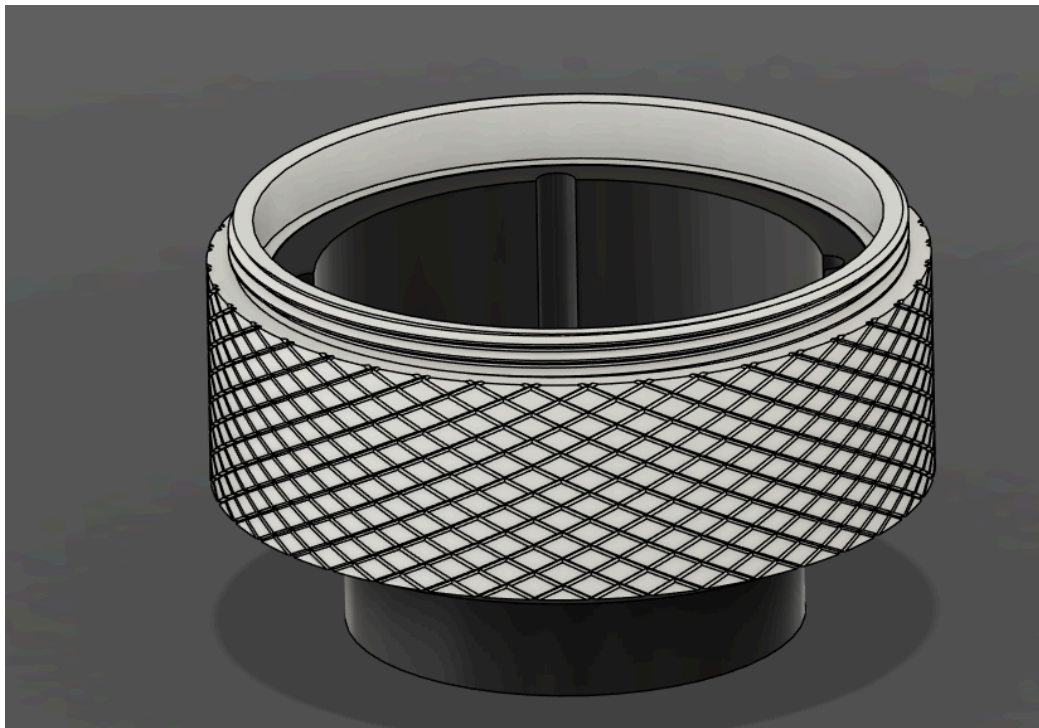
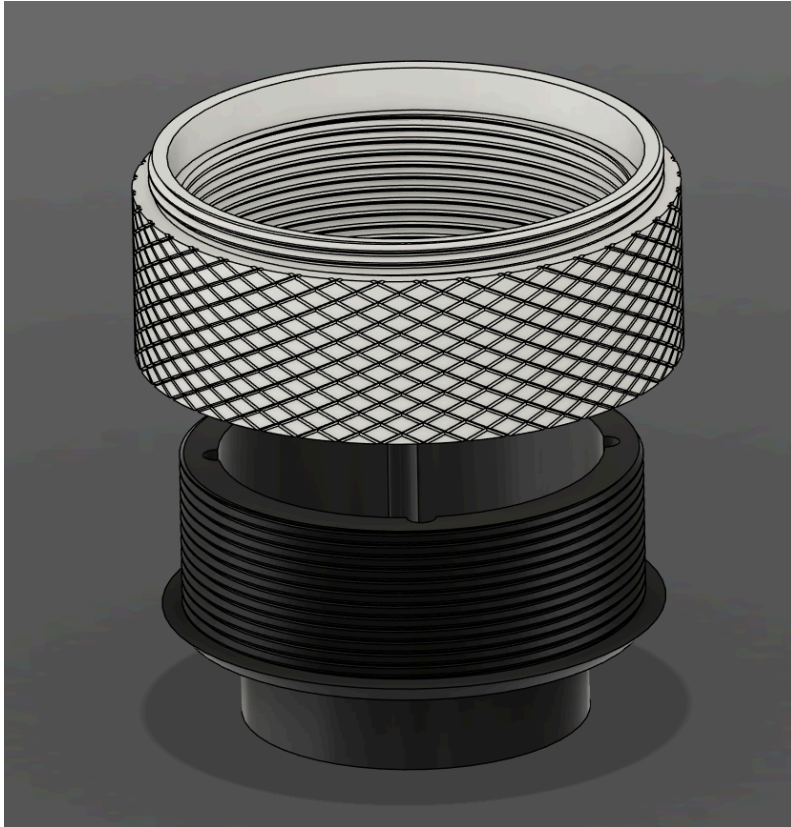


Part 2: Focuser/Lens Holder



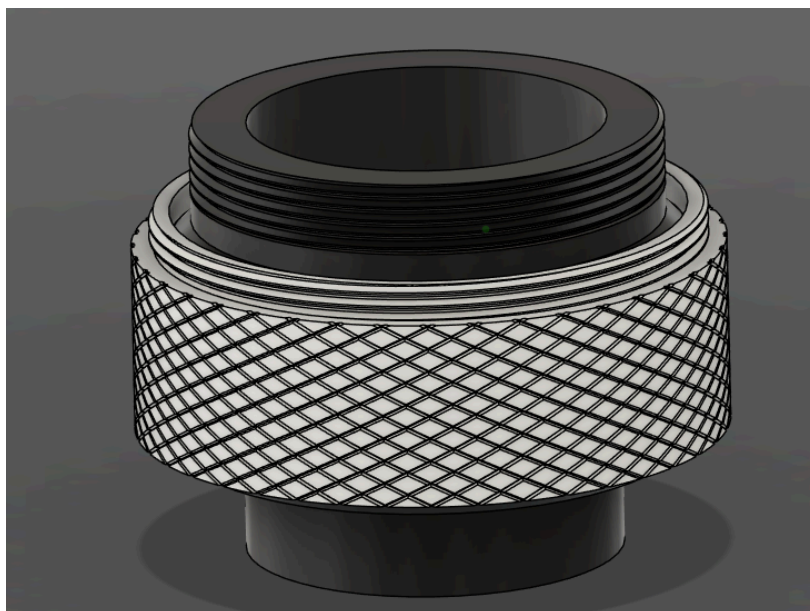
Step 1: Screw the Focus Knob onto the Base Flange/Thread

- Heavily recommended to grease this thread for smooth movement.



Step 2: Insert Focus Slider

- Be sure to align the bumps/grooves so that the slider moves in its track without rotating. Grease the mating cylindrical faces (the OD of the slider and ID of the Base Flange/Thread)



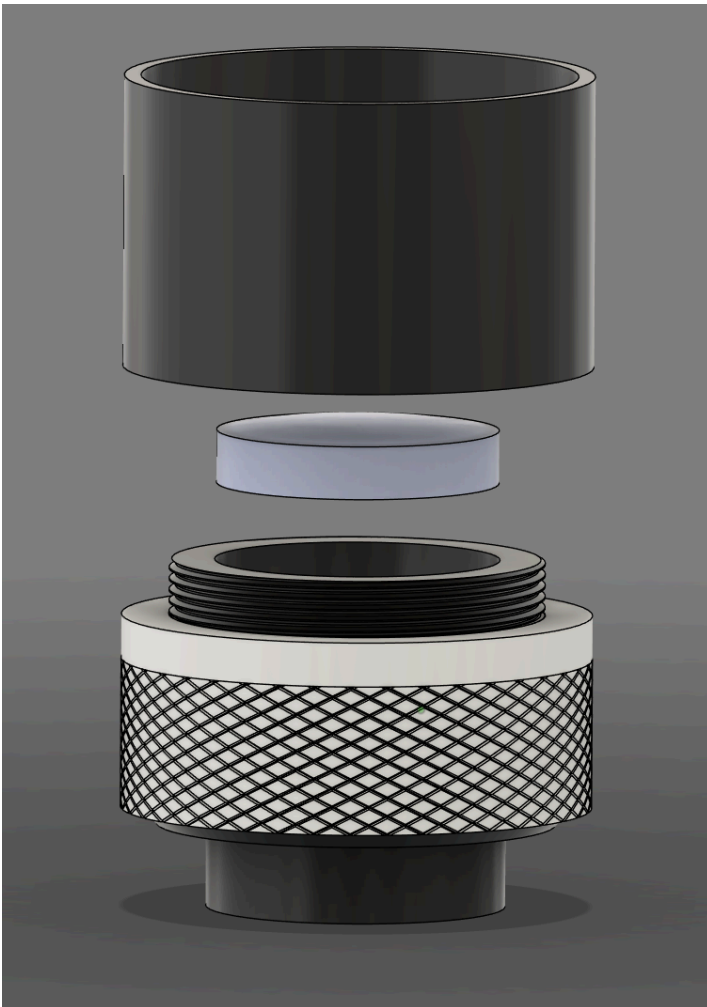
Step 3: Screw Retaining Ring onto Focus Knob

- No grease here. In fact, you could add glue if you want. This should not come loose. Make sure to check the Focus Knob can still spin and the Slider can slide.



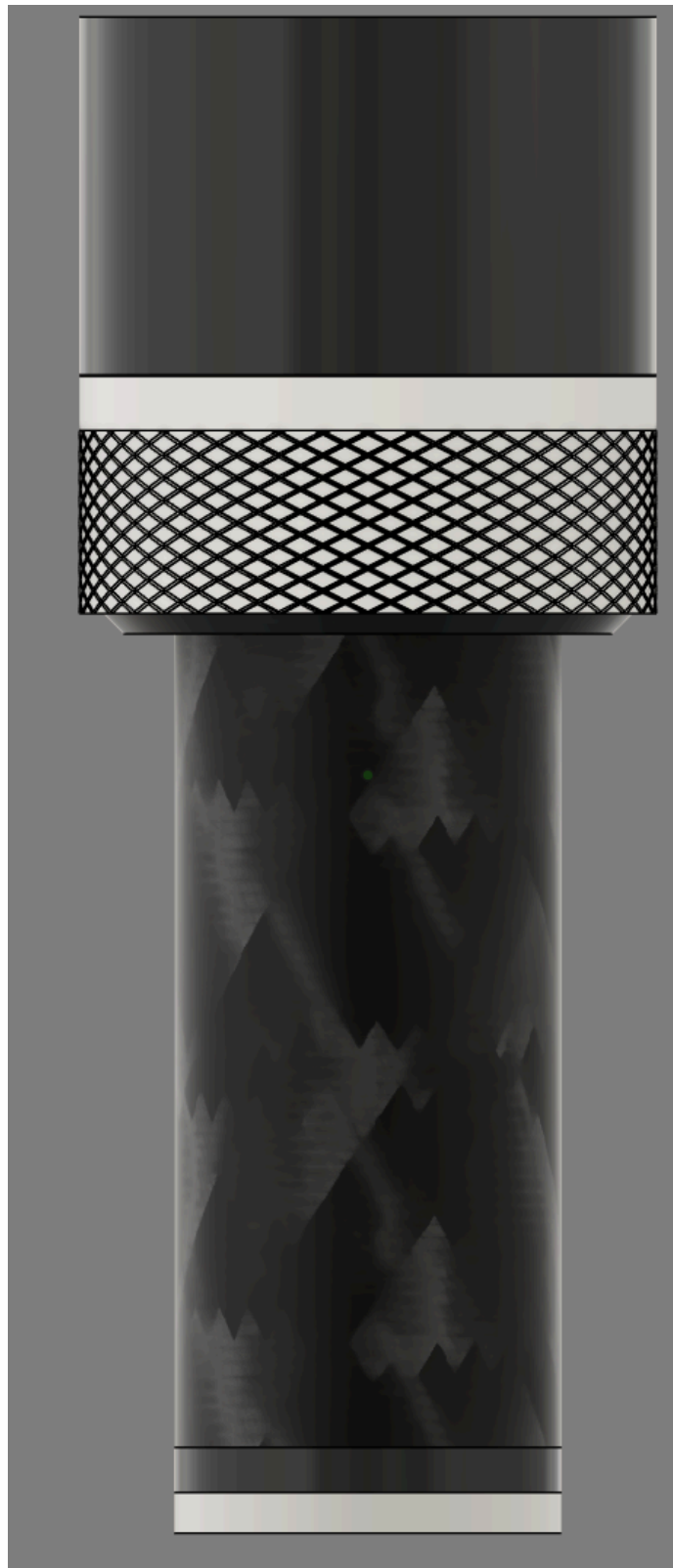
Step 4: Insert Lens and Screw On Holder/Dew Cover

- Insert the lens into the Focus Slider and then attach the Lens Holder/Dew Cover using the threads on the Focus Slider. These threads may also be glued. After tightening, ensure the Focus Knob can still spin and the Focus Slider can still slide. There is a lip inside the Focus Slider for the lens to rest on.



- The focuser/lens assembly is now complete.

Part 3: Bonding Parts Together



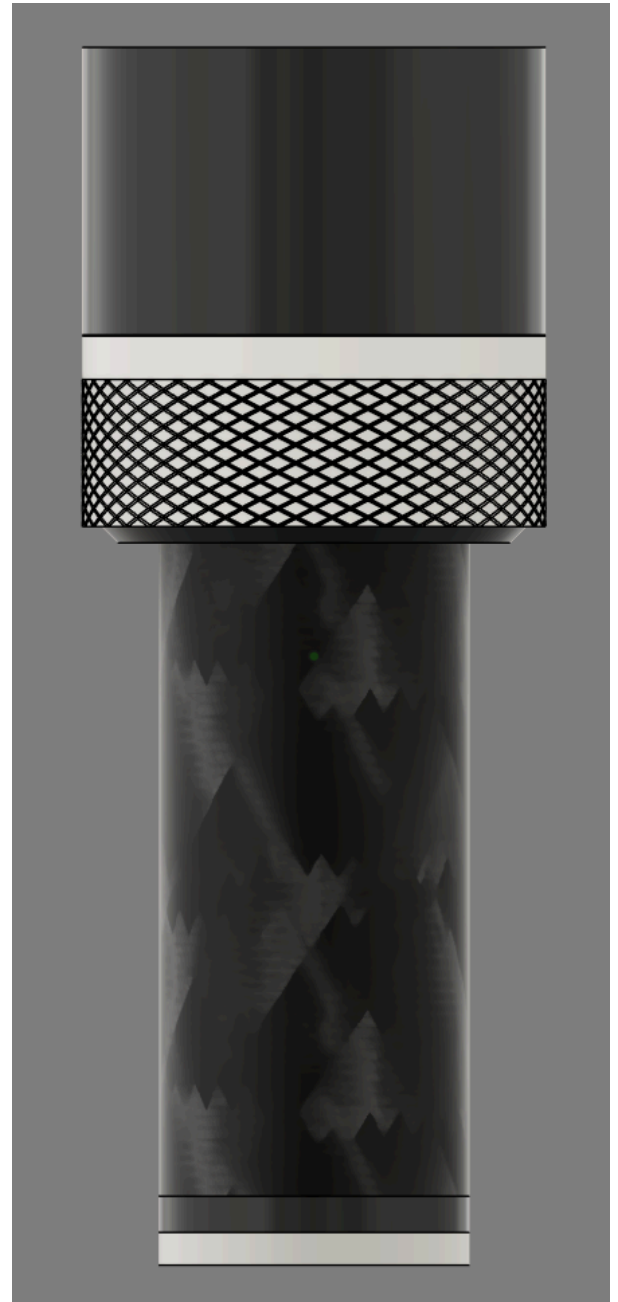
Step 1: Bond Camera Module into Carbon Fiber Tube

- The Camera Module has a lip on it. This lip is for interfacing with the 54mm ID Carbon Fiber Tube. Apply epoxy on this lip and then insert the Camera Module into the rear of the 54mm ID Carbon Fiber Tube



Step 2: Bond Focuser/Lens Module into Carbon Fiber Tube

- The Focuser/Lens Module has a lip on it. This lip is for interfacing with the 54mm ID Carbon Fiber Tube. Apply epoxy on this lip and then insert the Focuser/Lens Module into the rear of the 54mm ID Carbon Fiber Tube



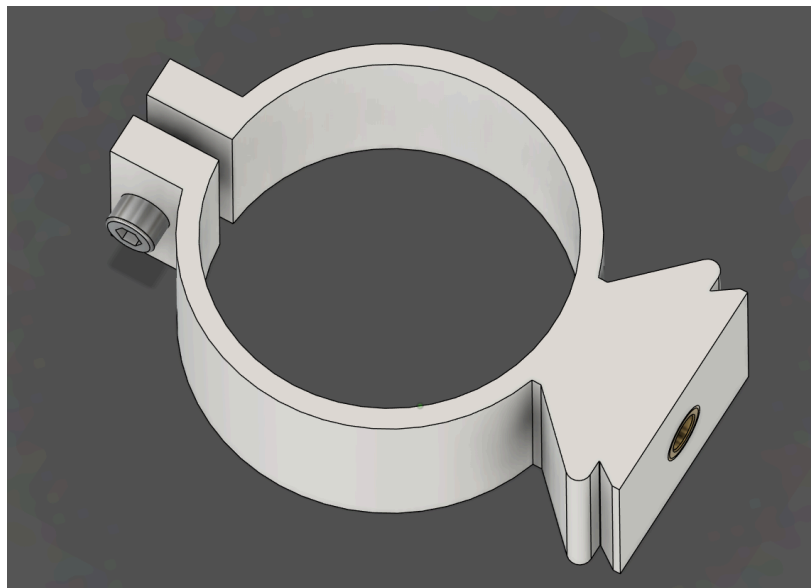
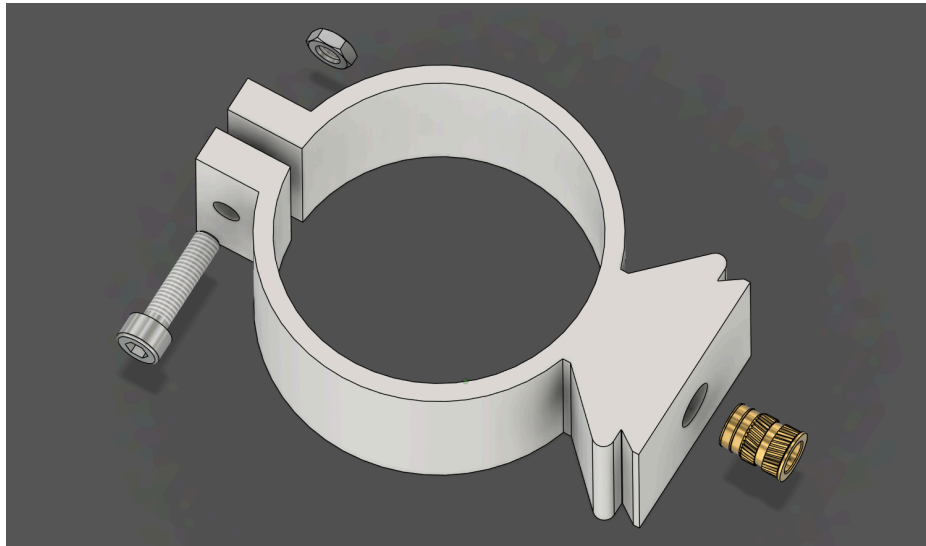
- The Scope is now Complete

Part 3: Mounting



Step 1: Pick Option

- There are 2 mounting clamp options provided. One has a Mini Vixen rail on it, common on Guider scopes, while the other has an Arcaswiss rail. Both also sport M6 mounting holes via heat insert. For whichever option you choose, you must insert the M6 heat insert, an M5 nut, and an M5x20 Bolt.



With That, Your SpaceMount Guider Scope is Now Complete!